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# Choosing EMG parameters: comparison of different onset determination algorithms and EMG integrals in a joint stability study

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**Abstract**

*Objective*. The aim was to test various algorithms for onset determination and compare onset repeatability to that from integrals of the EMG signal. The information contained in both parameters is discussed.

*Design*. Onset times were calculated using six different algorithms. The integral of the EMG signal was calculated for seven intervals: from tilt start and from each of the resulting onsets to 200 ms after tilt start.

*Background*. EMG onset times are often utilised, especially regarding co-ordination patterns or joint stability. There are almost as many different procedures for onset determination as authors dealing with it. Results in the literature are contradictory. The determination and usage of onset times remains controversial.

*Methods*. EMG signals from six muscles of the lower leg of 23 subjects were recorded during three consecutive, expected and unexpected sudden inversion and eversion trials while standing on a tilting platform.

*Results*. In most cases the repeatability of the onset times was considerably lower than that of the integrals of the EMG for all studied algorithms. In some cases earlier onset times corresponded to lower integral values and constant onsets to variable integrals.

*Conclusions*. In many cases onset times alone are not sufficient for describing onset phenomena. The additional calculation of the integrated EMG might provide relevant information regarding the quality of early activation.

**Relevance**

The findings are evidencing that care should be taken when interpreting onset times alone. The additional use of the integral of the EMG signal is suggested to provide more meaningful information.
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*Keywords:* Onset times; Integrated electromyography; Electromyography; Repeatability; Stability

## 1. Introduction

Muscle onset times are often utilised in studies concerned with electromyography (EMG), most of them regarding co-ordination patterns or joint stability. In the literature we find pure visual determination protocols (Ebig et al., 1997; Hodges and Bui, 1996), computer aided protocols at which the experimenter has to decide whether or not to accept the detected onset or where to finally place it (Hodges and Bui, 1996; Di Fabio, 1987), as well as fully automated algorithms (see Table 1). Different filters and onset thresholds are used and as stated by Hodges and Bui (1996) in many cases the criteria or methods for onset determination are not even specified. Sometimes a fixed value is used as threshold (Zhou et al., 1995), but in most algorithms the threshold is based on the standard deviation or a percentage of the EMG signal while the corresponding muscle is relaxed. As indicated by Winter (1984), the chosen threshold strongly influences the instant of onset detection. Further, the quality of onset time detection strongly depends on the quality of the signal. The fact of having a broad variety of methods (Table 1) and the lack of consensus might indicate that none of the known techniques is really satisfactory nor generally accepted. Benesch et al. (2000) despite stating that peroneal reaction time is a stable and reliable

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Table 1
Several authors and the corresponding methods for onset determination used

| Author (year) | Onset determination |
|---|---|
| Tomberg et al. (1991) | Electronic determination by a manually adjusted threshold |
| McKinley and Pedotti (1992) | At above 95% confidence interval for baseline during more than 10 ms |
| Johnson and Johnson (1993) | At 200% above noise |
| Zhou et al. (1995) | At 0.015 mV above baseline value |
| Lynch et al. (1996) | 10 SD above lowest rest level in several 100 ms RMS of 200 ms resting EMG |
| Ebig et al. (1997) | Visual determination |
| Bonato et al. (1988) | A statistical method using a double threshold detector |
| Duncan and McDonagh (2000) | Average of the 10 highest data points in a fixed window (35–80 ms) |
| Vaes et al. (2001) | At amplitude twice the peak to peak amplitude of average signal noise |

parameter, further state that its clinical relevance is not yet clear. So the determination and usage of onset times remains a controversial topic. It is very difficult to differentiate the real variability of a parameter from the variability due to flaws in the detection algorithm (Tomberg et al., 1991). However, making some assumptions, it is possible to estimate the contribution of different sources to the total variability (Brinckmann et al., 2002). Bonato et al. (1998) developed a statistical method using a double threshold detector. This method yields standard deviations below 15 ms when analysing gait patterns. This is enough for many applications and may reflect a real variability. However it is still too much for many other applications, e.g. studies dealing with EMG latency times in response to sudden tilts, where the reported differences between groups are often in the range or below this standard deviation.

The integrated EMG (iEMG) is frequently used for describing EMG activity. There is little, if any, controversy about its use. Further, many studies have dealt with its reliability (Gollhofer et al., 1990; Mero and Komi, 1986; Goodwin et al., 1999; Yang and Winter, 1983). Most of them reporting high correlation values for repeated measures.

To be useful, a parameter should provide relevant information and should be reliable. Onset times have been used for analysing proprioception, but when studying joint stability, the amount of early activity might be crucial and not always related to the onset time. It is hypothesised that: (a) onset time detection produces very variable results whereas the calculation of iEMG provides a higher repeatability and (b) onset times and iEMG are not necessarily related to each other. Therefore the aims of this study were: (a) to test the reproducibility of several onset determination algorithms and compare it to that from the iEMG and (b) expose and compare the information provided by these parameters.

## 2. Methods

The EMG signals (1000 Hz) from 23 subjects (12 males, 11 females; 13 stable, 10 unstable) were recorded using bipolar, pre-amplified surface EMG electrodes placed over the belly of six muscles of the lower leg (*peroneus longus, peroneus brevis, soleus, gastrocnemius lateralis, gastrocnemius medialis and tibialis anterior*) with an interelectrode distance of 2 cm. The grouping criterion was the subjective feeling of ankle/foot stability as determined by a short anamnestic questionnaire including among others frequency, kind and consequences of inversion trauma. Stable and unstable subjects were analysed separately, since stability could affect the results (Lynch et al., 1996; Vaes et al., 2001). The subjects underwent expected and unexpected sudden inversion and eversion trials (20°) while standing on a tilting platform. The subjects had bare feet and were aware of the tilting direction. Their left leg was full weight bearing, having its longitudinal axis parallel to the tilt axis of the plate. The tip of the right foot rested on a block to help maintain balance (Fig. 1). The subjects were instructed to look forward to a spot on the wall. The plate was released from behind the subjects out of their field of view. Each subject performed three trials per condition. The instant of release was indicated by counting backwards 3, 2, 1, tilt. At those trials randomly chosen to be unexpected the plate was released at any time during the countdown. The first trial of each condition was always unexpected. The onset times were determined using an algorithm with six different parameter combinations (Table 2). It was defined as the time between start of the tilt, as determined by an

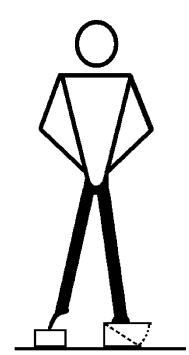

Fig. 1. Positioning of the subject on the tilt platform. Left foot was full weight bearing having its longitudinal axis parallel to the tilt axis of the plate. The tip of the right foot rested on a block to help maintain balance.

Table 2
The onset times were determined using two filters and three thresholds: Mean plus 2, 3 and 4 standard deviations (SD)

| Median filter, window width 13 points | | | Median filter, window width 26 points | | |
|---|---|---|---|---|---|
| Onset 1 at 2 SD | Onset 2 at 3 SD | Onset 3 at 4 SD | Onset 4 at 2 SD | Onset 5 at 3 SD | Onset 6 at 4 SD |

electrogoniometer (1000 Hz) aligned with the axis of rotation of the tilt plate, and the instant at which the filtered signal exceeded the given threshold. Two different median filters, window widths of 13 and 26 data points respectively, were applied to the rectified EMG signal (Fig. 2), since filtering may affect the onset calculation and as shown by Kadaba et al. (1985) also the repeatability of EMG phasic activity. As the detection threshold has shown to be a major factor influencing onset time detection (Winter, 1984) we tested three different thresholds, namely the mean plus 2, 3 or 4 standard deviations of the raw rectified activity, recorded prior to the tilting. For iEMG calculation the EMG signals were rectified and smoothed using a second-order Butterworth filter with a cut-off frequency of 10 Hz. The filtered data were normalised as follows (Arampatzis et al., 2001).

$$\mathrm{EMG}_{Nk} = \frac{\mathrm{EMG}_{Fk}}{\mathrm{Max}_k} \cdot 100$$

where $\mathrm{EMG}_{Nk}$ is the normalized EMG-data from $k$-muscle, $\mathrm{EMG}_{Fk}$ is the linear envelope EMG-data from $k$-muscle, $\mathrm{Max}_k$ is the maximal amplitude of the smoothed signal from $k$-muscle of each subject during the first lateral tilt.

Then the integrals were calculated from start of the tilt (Int.-Total) and from each of the six determined onset times to 200 ms after start of the tilt (Integrals 1 to 6).

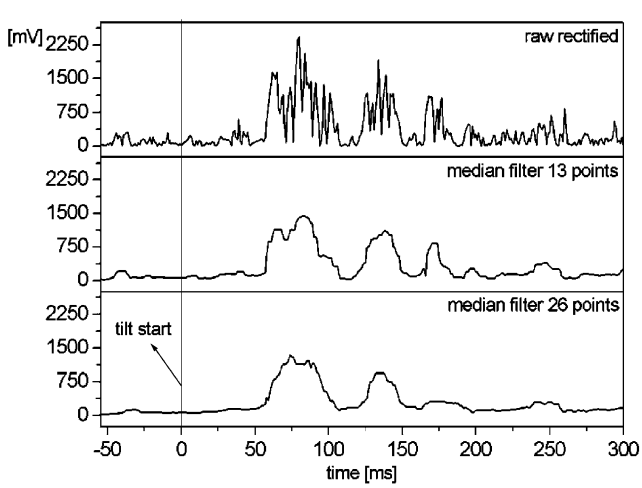


Fig. 2. Rectified and filtered (13 and 26 points median filter) EMG signal of the *m. peroneus longus* during a lateral tilt trial.

The data was split into eight groups resulting from the combination of the three dichotomic categories: stability, expectation and tilt direction. To verify the repeatability of the obtained onset times and integrals we first calculated the intraclass correlation coefficients (ICC) of all parameters for all three trials from each subject in every condition to asses the linearity of their relationship. Afterwards the Friedman test (a non-parametric test for $k$ dependent samples) was applied to check for possible differences ($P < 0.05$) within the set of correlated trials (Yang and Winter, 1983; Kamen and Caldwell, 1996). Finally, to give an idea of the absolute difference between trials, the root mean square differences (RMS), were also calculated.

As an estimator of the sources of variability, the variance due to biovariability and that due to the measurement were estimated as suggested by Brinckmann et al. (2002). The coefficient of variance (CV) was then calculated as the square root of the corresponding variance divided by the mean.

$$\sigma^2_{\mathrm{bio}} = R \cdot \sigma^2_{\mathrm{total}}$$

$$\sigma^2_{\mathrm{measurement}} = (1 - R) \cdot \sigma^2_{\mathrm{total}}$$

$$\mathrm{CV} = \frac{\sigma}{\bar{x}}$$

where $\sigma^2_{\mathrm{bio}}$ is the variance due to biovariability, $\sigma^2_{\mathrm{measurement}}$ is the variance due to the measurement, $\sigma^2_{\mathrm{total}}$ is the variance of the measured values, $R$ is the Pearson's correlation coefficient and $\bar{x}$ is the mean of the parameter values.

## 3. Results

An ICC above 0.7 was considered to correspond to an acceptable correlation. It is assumed that when no significant differences ($P < 0.05$) between correlated trials is found, a higher amount of ICCs above 0.7 corresponds to a better repeatability. As this does not inform about the magnitude of the deviation, the RMS values are also reported. As onset times and integrals have different units, to allow a comparison between these parameters, the RMS values are additionally expressed as a percentage of the corresponding means.

The Friedman test only revealed significant differences ($P < 0.05$) between consecutive measures in 2%, 3% and 6% of the cases for onset times (Onset 1–6), integrals from onset to 200 ms after tilt (Integral 1–6) and integrals from tilt to 200 ms after (Int.-Total), respectively. Since the results are focusing on repeatability, when citing amounts or percentages of ICCs above certain level, those trials displaying differences ($P < 0.05$) between consecutive trials are not included. In order to provide some more information, in Table 3 the results are split into more levels of correlation.

Table 3
Percentage of cases displaying no differences ($P < 0.05$) between correlated trials and having an ICC above given values for the different trial conditions

| | St-Ux-In | | St-Ex-In | | St-Ux-Ev | | St-Ex-Ev | | Un-Ux-In | | Un-Ex-In | | Un-Ux-Ev | | Un-Ex-Ev | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Ons. | Int. | Ons. | Int. | Ons. | Int. | Ons. | Int. | Ons. | Int. | Ons. | Int. | Ons. | Int. | Ons. | Int. |
| >0.7 | 31 | 67 | 25 | 86 | 39 | 78 | 31 | 75 | 19 | 69 | 50 | 72 | 42 | 89 | 53 | 69 |
| >0.8 | 8 | 17 | 8 | 42 | 14 | 72 | 19 | 61 | 17 | 58 | 39 | 50 | 22 | 50 | 39 | 47 |
| >0.9 | 0 | 0 | 6 | 8 | 3 | 14 | 6 | 33 | 6 | 8 | 19 | 11 | 14 | 14 | 28 | 19 |

Data include all muscles and algorithms. Stable (St)/Unstable (Un)—Unexpected (Ux)/Expected (Ex)—Inversion (In)/Eversion (Ev); Ons. = Onset times/Int. = Integrals.

Table 4
Average root mean square differences (RMS) between trials for the different trial conditions

| | St-Ux-In | St-Ex-In | St-Ux-Ev | St-Ex-Ev | Un-Ux-In | Un-Ex-In | Un-Ux-Ev | Un-Ex-Ev |
|---|---|---|---|---|---|---|---|---|
| Onset [ms] | 22 | 29 | 24 | 26 | 20 | 25 | 23 | 20 |
| Int. [%s] | 2.40 | 2.49 | 3.91 | 2.76 | 1.87 | 3.06 | 1.91 | 1.92 |
| Onset [%] | 42 | 79 | 45 | 70 | 43 | 57 | 49 | 62 |
| Int. [%] | 32 | 34 | 47 | 35 | 32 | 50 | 40 | 40 |

Absolute values and percentages of the mean. Data include all muscles and algorithms. ST (Stable)/UN (Unstable)—Ux (Unexpected)/Ex (Expected)—In (Inversion)/Ev (Eversion); Onset = Onset times/Int. = Integrals.

Table 5
Average root mean square (RMS) differences between trials for the different muscles

| | PL | PB | SOL | GL | GM | TA | Mean |
|---|---|---|---|---|---|---|---|
| Onset [ms] | 23 | 25 | 27 | 27 | 30 | 24 | 26 |
| Int. [%s] | 1.74 | 1.83 | 2.32 | 2.85 | 3.92 | 5.10 | 2.96 |
| Onset [%] | 52 | 73 | 51 | 48 | 50 | 73 | 58 |
| Int. [%] | 36 | 32 | 32 | 45 | 59 | 64 | 45 |

Absolute values and percentages of the mean. Data include all groups and algorithms. PL (*m. peroneus longus*), PB (*m. peroneus brevis*), SOL (*m. soleus*), GL (*m. gastrocnemius lateralis*), GM (*m. gastrocnemius medialis*) and TA (*m. tibialis anterior*). Onset = Onset times/Int. = Integrals.

Onset times vs. integrals: For Onset 1–6, thirty-six percent of all cases have an ICC above 0.7. For Integral 1–6 and for Int.-Total the corresponding percentages are considerably higher, namely 73% and 81% respectively. The RMS percentages are also higher for the onsets in almost all conditions (Tables 4 and 5). At all other comparisons done below, the integrals still provide more repeatable results than onset times.

Comparing algorithms: Onset 1 displays the highest amount of ICCs above 0.7 (44%) and Onset 3 the lowest one (27%). Integral 6 has the highest number of ICCs above 0.7 (81%), whereas Integral 1 has the least (58%).

Comparing trial conditions: Unstable subjects seem to produce more repeatable onset times than stable subjects. No bigger differences are evident when looking at the integrals. Between trial conditions no further systematic differences regarding repeatability can be identified (Tables 3 and 4).

Comparing muscles: Fig. 3 should help to better illustrate the repeatability of the parameters regarding the different muscles. The onset times for *m. tibialis anterior* and *m. peroneus longus* show quite more high ICC values than those from the other muscles. Interestingly when looking at the integrals, the *m. tibialis anterior* displays the lowest amount of high ICCs together with *m. peroneus brevis* and *m. gastrocnemius medialis*. Table 5 presents the absolute and the relative RMS values for the different muscles. Fig. 3 and Table 5, besides a lower variability for integrals also indicate that there might be discrepancies between the variability of onset times and integrals.

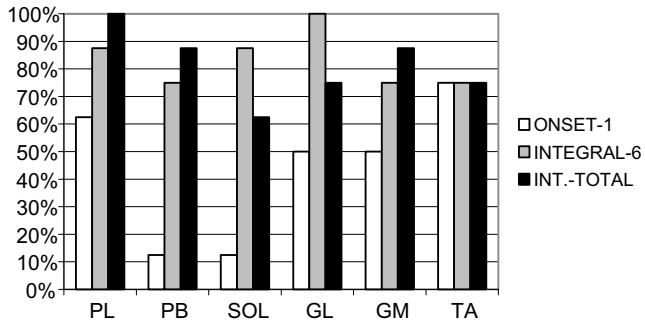


Fig. 3. Percentage of onset times and integrals demonstrating an ICC above 0.7 and no differences ($P < 0.05$) between trials for all studied muscles: PL (*m. peroneus longus*), PB (*m. peroneus brevis*), SOL (*m. soleus*), GL (*m. gastrocnemius lateralis*), GM (*m. gastrocnemius medialis*) and TA (*m. tibialis anterior*). Onset 1 and Integral 6 were chosen since they are showing highest number of ICCs above 0.7 for the respective parameter. In addition the integral from tilt start to 200 ms after (Int.-Total) is represented, for it is not dependent on the onset times.

Table 6
Means and standard deviations ( ) of the calculated onset times [ms] and integrals [%s] for all muscles and subjects

| Onset | | | | | | Integral | | | | | | Int.-Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 1 | 2 | 3 | 4 | 5 | 6 | |
| 38 (29) | 45 (31) | 50 (32) | 45 (32) | 52 (33) | 56 (33) | 6.62 (3.77) | 6.41 (3.77) | 6.30 (3.84) | 6.45 (3.90) | 6.35 (4.17) | 6.28 (4.14) | 7.15 (4.34) |

Algorithms 1–3 correspond to median filter window width 13 points; 2, 3 and 4 standard deviations respectively. Onset algorithms 4-6 correspond to median filter window width 26 points; 2, 3 and 4 standard deviations respectively. For the integrals the amplitude normalised EMG was integrated from the corresponding onset to 200 ms after tilt start. For Int.-Total the signal was integrated from tilt start to 200 ms after.

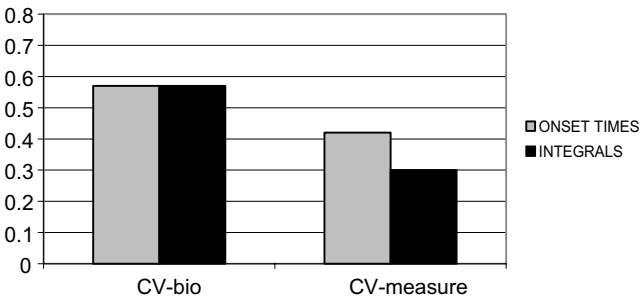


Fig. 4. Coefficients of variance due to biological variability ($CV_{bio}$) and due to the measurement ($CV_{measure}$), for all onset time algorithms (Onset 1–6) and for all calculated integrals (Integral 1–6 and Int.-Total).

Table 6 shows the means and standard deviations of the calculated onset times and the integrals for all muscles and subjects. It can be seen that Onset 1 has the earliest onset detection, followed by Onsets 2, 4, 3, 5 and finally Onset 6. The standard deviations are quite high and similar for all algorithms.

For some subjects, very early onset times together with very low iEMG were observed for some muscles, even when integrating from start of tilt to 200 ms after.

The magnitude of the variability as estimated by the RMS is quite high (Tables 4 and 5). Also The coefficient of variability due to biovariability is quite high (0.57) and identical for onset times and integrals. That of the measurement was 0.45 and 0.30 for onset times and integrals respectively (Fig. 4).

## 4. Discussion

Concerning the repeatability as estimated by the ICC when no differences ($P < 0.05$) between correlated trials is found, in most of the cases the integrals provide quite higher ICC values than onset times (Table 3 and Fig. 3). The higher variability of the onset times is also reflected in the higher RMS values expressed as percentage of the mean (Tables 4 and 5). None of the algorithms used, either for onset time determination or for the calculation of the integrals demonstrated a consistently high repeatability for all muscles or trial conditions. This lack in consistency may have several reasons:

(a) The interactions between the different conditions might differently affect the reproducibility (e.g. inversion trials producing more repeatable results for *m. peroneus longus* but less for *m. tibialis anterior*). So some algorithms might perform better for a given muscle and task combination but not for another. This is supported by our results and to some extent reflected in Fig. 3.
(b) The generally high variability of the results also favours a lack in consistency. As shown by the CV, the biological variability seems to play a major role (Fig. 4).
(c) Possibly surface EMG might not be accurate enough for this purpose. One reason might be that EMG is only estimating and not measuring muscle activation. Further, when using EMG only a portion and not the whole muscle is being measured and finally: Biopotentials measured at the skin are very low and despite being measurable, they are very sensitive to artifacts and interferences, which could lead to false onset detection.

The standard deviations in Table 6 are reflecting the intersubject variability, and therefore do not serve as criterion to judge the accuracy of the different algorithms. However, the intersubject variability is equally described by all algorithms since all of them display similar standard deviations.

Regarding the grouping and the trial conditions the only systematic observation was that unstable subjects seem to have a more constant behaviour regarding the onset times whereas no major differences can be found in the integrals. As the main criterion for subject selection was the subjective feeling of instability, unstable subjects might have been more alert than stable ones, this way achieving a higher reproducibility. Possibly other systematic differences between trial conditions could have been identified if we had taken a bigger sample of subjects and/or amount of repeated measures.

The review of the literature and the presented results indicate that the determination of onset times is critical. Our results show that the integrals during the different intervals considered are quite more repeatable than the onset times and that onset times have only a small effect on the integral itself. Stability studies dealing with EMG

focused their attention mainly on reaction times. Some of them find prolonged reaction times in unstable subjects (Konradsen and Ravn, 1990), others do not find any differences (Johnson and Johnson, 1993). For some subjects, very early onset times were observed for some muscles simultaneously having a very low iEMG: This, together with the difficulties inherent to the identification of onset times, might also lead to the controversial conclusions regarding onset times found in the literature and to the questioning of its clinical relevance. Late onset times have been associated to ankle instability (Konradsen and Ravn, 1990). Early but poor activation and later but higher activation could represent two different strategies. When relying only on onset times, it is not possible to distinguish between subjects displaying late onsets and low iEMG and those having the same late onsets but higher iEMG. First strategy could be hypothesised as risky whereas the second one could represent a safe alternate strategy to early and low activation.

## 5. Conclusion

Our results demonstrate that relying on only the onset times might be insufficient in many applications. Whenever we are interested in the amount of activity of a certain muscle, we have to use the iEMG or a related parameter. The onset times only provide an instant in time whose determination is critical and which is not necessarily related to the activity occurring afterwards. The iEMG estimates the amount of activity done by the muscle. By setting the integration interval around the expected onset time, useful information relevant to the quality of the onset can be obtained.

Considering the repeatability of both studied parameters and the information provided by them, it is suggested that conclusions drawn about joint stability basing only on the information contained in onset times have to be carefully considered. In many applications the combined use of iEMG and onset times is preferred to the use of onset times alone.

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